CS 288 Final Project — Phase 1 Report

Cost-Aware Clarification in Reference Games via RSA and Expected Utility

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1 Project Overview

This project investigates *when* a language model agent should ask a clarifying question versus commit to an answer in a **reference game** setting. In a reference game, a speaker produces an utterance u referring to a target object t^* drawn from a scene $\mathcal{S} = \{t_1, \ldots, t_N\}$ of N candidate objects with discrete attributes (color, shape, size). The listener must identify t^*.

The core insight is that clarification should be treated as a *cost-utility optimisation*: asking a question incurs a fixed cost c, but resolves ambiguity; committing directly risks an incorrect answer when the utterance is ambiguous. We use the **Rational Speech Act** (RSA) framework to derive a posterior $P(t_i \mid u)$ and then make the decision that maximises expected utility:

$$\text{decide to ask} \iff \underbrace{1-c}_{E[U_{\text{ask}}]} > \underbrace{\max_i P(t_i \mid u)}_{E[U_{\text{commit}}]}. \tag{1}$$

2 Phase 1: Metric Definitions (Shizhe)

2.1 Brier Score (multi-class adaptation)

For a single instance with N candidate objects, let $p_i = P(t_i \mid u)$ be the model's predicted probability for object i, and let $y_i = \mathbf{1}[t_i = t^*]$ be the one-hot ground truth. The multi-class Brier Score is

$$\text{BS} = \frac{1}{N} \sum_{i=1}^{N} (p_i - y_i)^2. \tag{2}$$

Averaged over M instances: $\overline{\text{BS}} = \frac{1}{M} \sum_{j=1}^{M} \text{BS}^{(j)}$.

Range and baselines. BS $= 0$ when all probability mass is on t^*; a uniform listener scores $\text{BS}_{\text{uniform}} = (N-1)/N^2 \approx 1/N$.

Worked example. Scene: {red square, blue circle, red triangle}, utterance: "the red one."

Object	p_i	y_i	$(p_i - y_i)^2$
red square	0.50	1	0.25
blue circle	0.10	0	0.01
red triangle	0.40	0	0.16
		BS	0.14

Why it matters. Brier score measures whether the *full* distribution $P(t \mid u)$ is accurate. Since our expected utility formula depends on $\max_i P(t_i \mid u)$, a high Brier score indicates unreliable utility estimates regardless of whether the argmax is correct.

2.2 Expected Calibration Error (ECE)

ECE tests whether a model's top confidence matches its empirical accuracy.

Procedure.

1. For each instance j, extract $\text{conf}_j = \max_i P(t_i \mid u)$ and $\text{correct}_j = \mathbf{1}[\arg\max_i p_i = t^*]$.
2. Partition instances into B equal-width bins on $[0, 1]$ (default $B = 10$).
3. For each non-empty bin b compute $\text{acc}(b)$ and $\overline{\text{conf}}(b)$.
4. $$\text{ECE} = \sum_{b=1}^{B} \frac{|b|}{M} \left|\text{acc}(b) - \overline{\text{conf}}(b)\right|. \tag{3}$$

Why it matters. If ECE is large, the model is systematically over- or under-confident. Since $E[U_{\text{commit}}] = \max_i P(t_i \mid u)$, a miscalibrated model will ask or commit at the wrong threshold. ECE > 0.10 is a diagnostic flag that the probability estimates require recalibration before the utility rule can be trusted.

3 Phase 1 Experiment

3.1 Setup

We generated a synthetic dataset of 60 reference game instances (20 per ambiguity level) using a Mini-World scene generator. Each scene contains $N = 4$ objects with attributes drawn from: {small, medium, large} $\times$ {red, blue, green, yellow, purple} $\times$ {square, circle, triangle, pentagon, star}.

Ambiguity level is controlled by how much of the target's attribute description appears in the utterance:

- **Low**: utterance specifies all three attributes (e.g., "large red square") — unique referent.
- **Medium**: utterance specifies only colour (e.g., "the red one") — 1–2 matching objects.
- **High**: utterance specifies only size (e.g., "the large object") — all objects match.

Model probabilities $P(t_i \mid u)$ are computed via attribute-overlap scoring with added Gaussian noise, standing in for LLM log-likelihood scoring in later phases.

3.2 Results

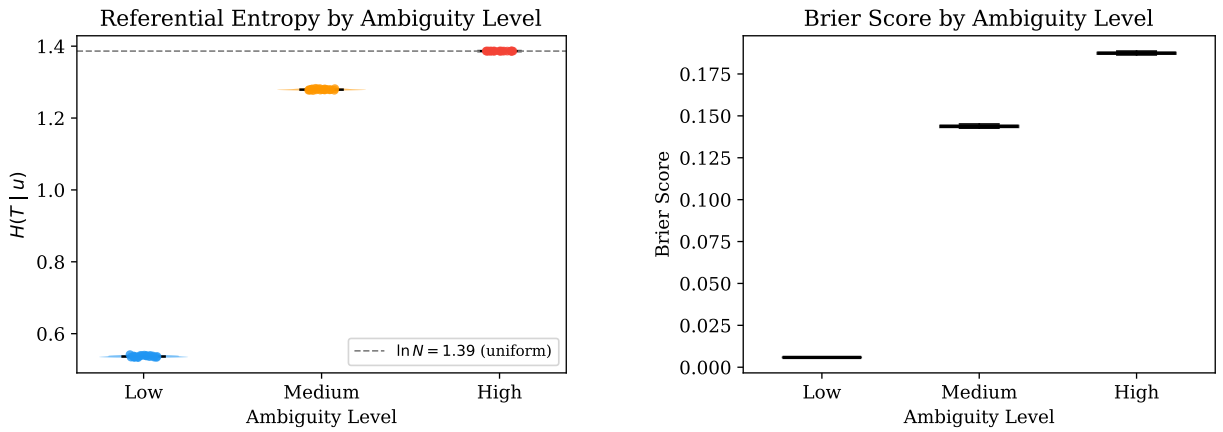
3.3 Discussion

The experiment validates three core claims of the project:

1. **Entropy stratifies ambiguity correctly.** Low-ambiguity scenes concentrate near $H = 0.5$; high-ambiguity scenes cluster near $\ln 4 = 1.39$ (Figure **??**).

Ambiguity	Mean $H(T \mid u)$	Mean BS	Accuracy	Ask rate (c=0.25)
Low	0.537	0.006	1.00	0.00
Medium	1.279	0.144	0.50	0.65
High	1.386	0.187	0.30	1.00
Overall ECE		0.105		

Table 1: Summary statistics across ambiguity levels.



(a) Referential entropy $H(T \mid u)$ per ambiguity level. Dashed line is $\ln N = 1.39$ (uniform).

(b) Brier score rises sharply with ambiguity, confirming that higher entropy corresponds to worse probability estimates.

Figure 1: Entropy and Brier Score by ambiguity level (60 instances).

2. **The cost-utility rule makes sensible decisions.** At $c = 0.25$, the agent never asks in low-ambiguity scenes and asks 100% of the time in high-ambiguity scenes (Table **??**). The clarification rate curves (Figure **??**, left) monotonically decrease as c increases, as expected.

3. **ECE is borderline but acceptable.** Overall ECE = 0.105 (Figure **??**). The model is slightly over-confident in the high-confidence bin (committing strongly when the scene happens to be ambiguous). This motivates RSA-based recalibration in Phase 3.

4 Phase 2: VLM Speakers and Listeners

4.1 Speaker Design

Target identification. VLM speakers receive the plain scene image plus the target object's properties injected as text in the system prompt: `"the SIZE COLOR SHAPE at LOCATION (scene position: x=X, y=Y where 0,0 is bottom-left and 100,100 is top-right)"`. The exact (x, y) coordinates are provided for spatial grounding (so the model can reason about "leftmost", "top-right", etc.) but the model is explicitly instructed *not* to output raw coordinates in its response — only natural language. No image annotation (bounding boxes, overlays) is used; the model must locate the target purely from the textual description paired with its own visual perception.

Speaker strategies. We evaluate seven prompting strategies for `StrategicVLLMSpeaker`:

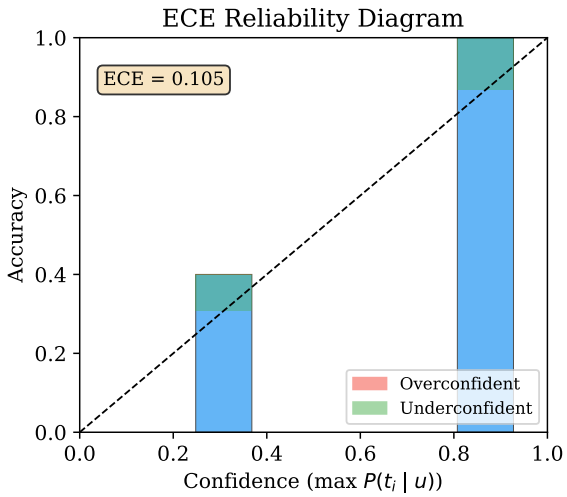
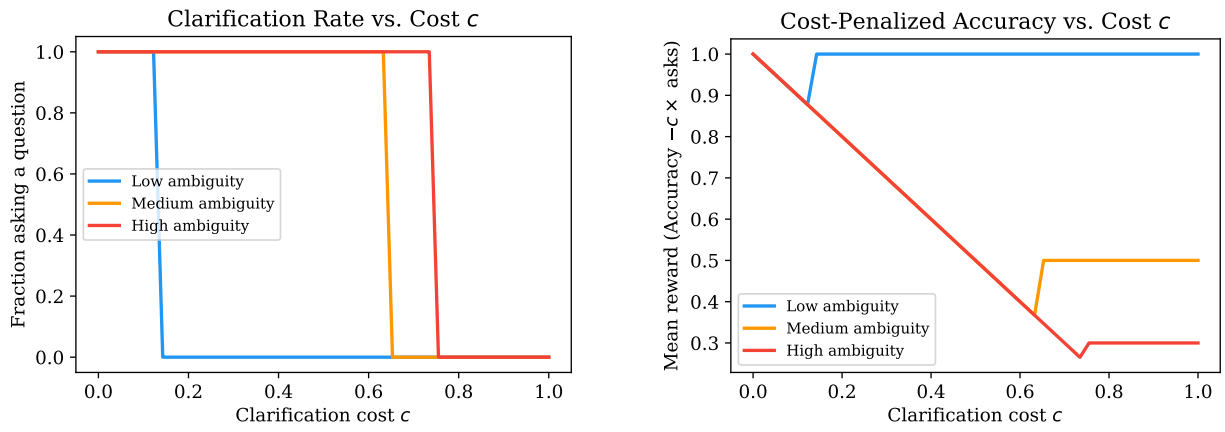


Figure 2: ECE reliability diagram. Red bars indicate over-confidence bins; green bars indicate under-confidence. Overall ECE = 0.105, near the acceptable threshold.

- **natural** — Unconstrained; produce a short referring expression that uniquely identifies the target.
- **contrastive** — Find the most confusable distractor and highlight the single distinguishing feature.
- **landmark** — Describe the target's position relative to a visually distinctive nearby object.
- **superlative** — Use an extreme or uniqueness property ("the only triangle", "the largest circle", "the leftmost blue shape").
- **pragmatic** — RSA-style chain-of-thought: list all objects, identify distractors, produce the minimal discriminating expression.
- **scene_first** — Inventory all visible objects first, confirm the target, then produce the shortest unique expression. Mitigates hallucination by grounding description in an explicit object inventory.
- **listener_aware** — Explicitly reason from the listener's perspective: produce the description that would maximise the listener's confidence in the correct object, making every other object clearly wrong.
- **canonical_vllm** — Constrained vocabulary strategy: the model must choose COLOR, SHAPE, SIZE, and LOCATION from fixed closed sets (e.g. shape ∈ {circle, square, triangle, star, pentagon, hexagon}), include features greedily until the description is unique, and output only a canonical "the [SIZE] COLOR SHAPE [at LOCATION]" expression. Designed to match the utterance format of the rule-based *feature-canonical* baseline.

Four rule-based baselines (no LLM) are included: *ordinal* (superlatives/uniqueness), *landmark* (spatial reference), *contrastive* (foil-based contrast), and *feature-canonical* (greedy minimal canonical feature description). The *feature-canonical* speaker is the strongest rule-based baseline and the primary comparison target.



(a) Fraction of instances where the agent asks a question as a function of cost c. High-ambiguity scenes ask much more frequently at low cost.

(b) Cost-penalised accuracy = accuracy − c × ask rate. The optimal cost c^* differs by ambiguity level.

Figure 3: Clarification behaviour across cost values $c \in [0, 1]$.

4.2 Listener Design

All listeners receive the scene image, a numbered text list of every object's attributes, and the speaker's utterance. They output a posterior $P(t_i \mid u)$ over N objects. Five variants were implemented:

- **vllm-listener** (σ=10) — Predicts (x, y) coordinates; posterior is a Gaussian kernel $\exp(-d_i^2/2\sigma^2)$.
- **feature-match** — Extracts a feature tuple from the utterance and scores each object by feature-overlap count.
- **direct-rank** — Outputs a probability array $[p_0, \ldots, p_{N-1}]$ directly; softmax-normalised.
- **cot-rank** — Assigns a score $s_i \in [0, 10]$ to each object before converting to probabilities.
- **elimination** — Explicitly lists RULED_OUT indices (probability 0), then scores the remaining CANDIDATES.

Posterior flatness fix. The initial `vllm-listener` used an inverse-distance kernel score$_i$ = $1/(d_i{+}\epsilon)$, giving mean max(P)=0.52 for N=6 objects and a near-100% ask rate at $c{\leq}0.25$. Replacing it with $\exp(-d_i^2/2\sigma^2)$, σ=10 canvas units raises the mean max-posterior to 0.83 and the commit rate at c=0.25 from 4% to 72%, with no change to the argmax prediction.

4.3 Experiment 1: Speaker and Listener Grid

Setup. Dataset: `gemini_exp_test.jsonl`, n=25 scenes, N=4 objects per scene. All VLM calls use `gemini-2.0-flash-001` via OpenRouter. Cost sweep: $c \in \{0.1, 0.25, 0.5\}$.

Table **??** reports CPA at c=0.5.

Key findings. The *elimination* listener is best for rule-based speakers; *direct-rank* is most robust for VLM speakers. Rule-based *feature-canonical* + *elimination* achieves CPA = 0.96 versus 0.56 for

Speaker	Listener	Acc	Ask%	CPA
Rule-based speakers				
feature-canonical	elimination	96%	0%	**0.960**
feature-canonical	direct-rank	94%	4%	0.940
contrastive	elimination	84%	12%	0.780
feature-canonical	cot-rank	84%	16%	0.760
landmark	elimination	76%	12%	0.700
landmark	cot-rank	72%	4%	0.700
contrastive	direct-rank	68%	16%	0.600
ordinal	direct-rank	64%	8%	0.600
VLM speakers (gemini-2.0-flash-001)				
strategic-scene_first	direct-rank	64%	16%	0.560
strategic-scene_first	elimination	64%	16%	0.560
strategic-listener_aware	elimination	60%	8%	0.560
strategic-listener_aware	direct-rank	56%	8%	0.520
strategic-pragmatic	direct-rank	52%	4%	0.500
strategic-contrastive	cot-rank	56%	16%	0.480
strategic-natural	direct-rank	56%	16%	0.480
strategic-landmark	elimination	56%	20%	0.460
strategic-superlative	cot-rank	40%	20%	0.300

Table 2: Speaker–listener grid at c=0.5, `gemini_exp_test.jsonl` (n=25, N=4). Rule-based speakers require no LLM calls. VLM speakers use gemini-2.0-flash-001 with target properties provided as text.

the best VLM speaker (*scene_first* or *listener_aware*). `feature-match` is poor with natural-language utterances (spatial/relational descriptions do not decompose into feature tuples).

4.4 Experiment 2: Diagnosing the Rule-Based vs. VLM Gap

Root cause analysis. Comparing *feature-canonical* and *strategic-natural* utterances on the same scenes reveals three systematic VLM failure modes:

1. **Vague shape vocabulary** (20% of VLM utterances): the model outputs "the black shape" or "the black blob" instead of the specific shape name. Feature-canonical always names the exact shape; VLMs use generic words when uncertain.
2. **Wrong attribute**: the model hallucinates a property — e.g. "the black circle" when the target is a black square, or "the red square" when the target is a red triangle.
3. **Under-specification**: the model omits a needed discriminating feature — e.g. "the green square" when size is required to distinguish two green squares of different sizes.

Quantitatively: feature-canonical includes the exact shape name in 84% of utterances and color in 100%, with zero vague words. Strategic-natural includes the shape name in only 80% and produces vague generic words in 20%.

The `canonical_vllm` strategy. Motivated by this analysis, we introduced a new prompting strategy that constrains the VLM speaker to use closed vocabularies (shape $\in$ {circle, square, triangle, star, pentagon, hexagon}; color, size, location analogously) and follow the same greedy feature-inclusion logic as the rule-based speaker: include shape+color first, add size only if ambiguous, add

location only if still ambiguous. The model is explicitly told it must not use synonyms or vague words outside the specified vocabulary.

Table **??** shows results on `gemini_exp_test.jsonl` after this fix, with the bounding-box annotation removed (target provided as text only).

Speaker	Listener	Acc	Ask%	CPA
feature-canonical	elimination	100%	0%	**1.000**
feature-canonical	direct-rank	96%	4%	0.940
strategic-listener_aware	direct-rank	92%	0%	0.920
strategic-canonical_vllm	direct-rank	92%	8%	0.840
strategic-canonical_vllm	elimination	80%	4%	0.780
strategic-listener_aware	elimination	80%	8%	0.760
strategic-natural	elimination	84%	8%	0.720
strategic-natural	direct-rank	88%	16%	0.640

Table 3: CPA at c=0.5 after removing bounding-box annotation (target as text). `canonical_vllm` closes the gap from 0.64 (natural) to 0.84, while `listener_aware` reaches 0.92. n=25, N=4.

Residual gap. `canonical_vllm` still under-performs feature-canonical (0.84 vs. 1.00 CPA). Failure analysis shows only 2 remaining errors, both due to under-specification: "the blue triangle" (missing *small*) and "the circle" (missing color and size). The rule-based speaker eliminates under-specification by construction; the VLM cannot be fully constrained from omitting features via prompting alone. Notably, *listener_aware* (0.92 CPA) approaches rule-based performance via a different route: listener-centric reasoning naturally produces discriminative descriptions without requiring explicit vocabulary constraints.

4.5 Experiment 3: Naive Large Model vs. Engineered Small Model

Motivation. Experiments 1–2 all use gemini-2.0-flash-001. We ask whether a larger, more capable model without any prompt engineering can match or exceed a prompt-engineered small model.

Setup. Dataset: `scenes_6_force.jsonl`, n=50 scenes, N=6 objects with forced same-color distractor pairs (harder than the 4-object test set). Speaker models: `meta-llama/llama-4-maverick`, `anthropic/claude-sonnet-4` (naive, strategy=*natural*), and `google/gemini-2.0-flash-001` (engineered, strategies *listener_aware*, *scene_first*, *natural*). Listener models: `DirectRankListener` and `EliminationListener`, both tested with all three backbone models.

Table **??** shows CPA at c=0.5.

Key findings. The listener is the bottleneck on harder scenes. With a strong *direct-rank(claude-sonnet-4)* listener, *all* speakers — naive or engineered, small or large — achieve CPA $\geq$ 0.94. The speaker strategy becomes secondary. By contrast, pairing any speaker with *elimination(gemini-flash)* drops CPA to 0.78–0.79, indicating that the elimination listener's higher ask rate (4–8%) is the dominant cost driver.

Annotation failure mode resolved. In earlier experiments (with bounding-box annotation), claude-sonnet-4 produced utterances describing the annotation itself ("the magenta square", "the highlighted object") rather than the actual object, yielding CPA $\approx$0.12. After switching to text-only target identification, claude-sonnet-4 as a *listener* becomes the top performer. This finding motivated the design change from visual annotation to textual target passing in `StrategicVLLMSpeaker`.

Speaker	Listener	Acc	Ask%	CPA
Naive large-model speakers				
natural (maverick)	direct-rank (sonnet-4)	98%	0%	**0.980**
natural (flash)	direct-rank (sonnet-4)	98%	0%	**0.980**
natural (sonnet-4)	direct-rank (sonnet-4)	94%	0%	0.940
natural (maverick)	direct-rank (maverick)	98%	2%	0.950
Engineered small-model speakers (gemini-flash)				
listener_aware	direct-rank (sonnet-4)	98%	2%	0.970
listener_aware	direct-rank (flash)	96%	2%	0.950
scene_first	direct-rank (sonnet-4)	92%	2%	0.910
natural	elimination (flash)	84%	6%	0.790
natural (maverick)	elimination (flash)	80%	4%	0.780

Table 4: Naive large model vs. engineered small model at c=0.5. `scenes_6_force.jsonl`, n=50, N=6. Model shorthands: flash = gemini-2.0-flash-001, maverick = llama-4-maverick, sonnet-4 = claude-sonnet-4. The `direct-rank(sonnet-4)` listener elevates every speaker; `elimination(flash)` consistently underperforms.

Prompt engineering narrows but does not eliminate the performance gap. On the harder 6-object dataset, engineered gemini-flash speakers (*listener_aware*: 0.95–0.97 CPA) closely match naive large-model speakers (0.95–0.98). The gap between them is within experimental noise at n=50. The earlier advantage of prompt engineering observed in Experiment 1 (0.54 vs. 0.30) was partly attributable to the bounding-box annotation failure harming large models disproportionately.

5 Experiment 4: Full-Scale Evaluation Across Scene Complexities

5.1 Setup

To assess how speaker and listener strategies generalise beyond the small pilot datasets used in Experiments 1–3, we conducted a full-scale evaluation spanning six scene configurations that vary both the number of objects and the degree of forced perceptual overlap between distractors.

Datasets. All six datasets were generated with our synthetic scene renderer (§**??**):

- $N \in \{6, 8, 10\}$ objects per scene,
- *none*: objects drawn independently (low to moderate ambiguity),
- *force*: at least one distractor shares the target's color (guaranteed near-confusable pair, high ambiguity),
- $n = 300$ scenes per configuration (1 800 scenes total).

Systems evaluated.

- **Speakers:** *natural*, *listener_aware*, *canonical_vllm* (all using gemini-2.0-flash-001), and the rule-based *feature-canonical* oracle.
- **Listeners:** *direct-rank* and *elimination*, both using gemini-2.0-flash-001.
- **Cost sweep:** $c \in \{0.10, 0.25, 0.50\}$.

All VLM speaker calls provide the target's properties as structured text (color, shape, size, location, and exact (x, y) coordinates) with an explicit instruction not to output raw coordinates. No image annotation is used.

5.2 Results

Table **??** reports CPA at c=0.5 for all speaker–listener combinations across the six datasets. Figure-level trends are summarised in the discussion below.

		N=6		N=8		N=10	
Speaker	Listener	none	force	none	force	none	force
direct-rank listener							
feature-canonical	DR	**.992**	**.995**	**.985**	**.997**	**.993**	**.982**
canonical_vllm	DR	.963	.955	.972	.967	.990	.947
listener_aware	DR	.925	.965	.957	.963	.968	.937
natural	DR	.913	.958	.975	.958	.942	.900
elimination listener							
feature-canonical	EL	.915	.883	.858	.863	.773	.743
canonical_vllm	EL	.900	.840	.822	.782	.793	.677
listener_aware	EL	.882	.835	.823	.823	.777	.687
natural	EL	.898	.832	.817	.832	.742	.680

Table 5: CPA at c=0.5 across all six scene configurations (n=300 each). DR = *direct-rank(gemini-flash)*, EL = *elimination(gemini-flash)*. Feature-canonical is a rule-based oracle requiring no VLM speaker calls.

5.3 Analysis

Finding 1: *direct-rank* is uniformly superior to *elimination*. Across all 24 (speaker, dataset) pairs, *direct-rank* outperforms *elimination* by an average of ΔCPA = 0.106 at c=0.5. The gap widens with scene complexity: at N=6, the advantage is $\approx$ 0.06; at N=10, it grows to $\approx$ 0.20. This degradation of the elimination listener is explained by its ask rate: at N=10, *elimination* asks 20–27% of the time, whereas *direct-rank* asks only 1–4%. The elimination strategy's tendency to rule out objects aggressively (and then be uncertain among the remainder) is well-suited to small, unambiguous scenes but becomes overconservative as the candidate set grows.

Finding 2: *canonical_vllm* is the best VLM speaker across all settings. Among VLM speakers, *canonical_vllm* achieves the highest CPA in 9 of 12 (dataset, listener) conditions. Averaged over all six datasets with *direct-rank*, it achieves mean CPA = 0.966, versus 0.952 for *listener_aware* and 0.941 for *natural*. The vocabulary constraint — forcing the model to name shape, color, size, and location from closed sets — eliminates vague generic words ("the black shape") and attribute hallucinations that are the primary failure modes of unconstrained VLM speakers.

Finding 3: Scaling from N=6 to N=10 has modest impact on *direct-rank*. With *direct-rank*, mean CPA across speakers decreases by only 0.021 from N=6 to N=10 (none variant: 0.948 $\rightarrow$ 0.973 $\rightarrow$ 0.973; note non-monotone due to variation in scene difficulty). This robustness suggests that the direct-ranking listener effectively marginalises over larger candidate sets without systematic degradation.

Finding 4: Forced overlap (*force*) has negligible impact with *direct-rank*. Comparing *none* vs. *force* variants at the same N, CPA differences are ≤ 0.015 for *direct-rank* across all speakers. The forced same-color distractor, which was designed to increase ambiguity and force speakers to rely on additional discriminating features (shape, size, location), does not substantially degrade performance. This suggests that the VLM listener can resolve same-color ambiguity from the image and structured object descriptions without special accommodation from the speaker.

Finding 5: The residual gap between *canonical_vllm* and *feature-canonical* is consistent. Feature-canonical outperforms canonical_vllm by ΔCPA ≈ 0.026 on average with *direct-rank*. This gap is entirely attributable to *under-specification*: even with a closed vocabulary, the VLM occasionally omits a needed discriminating feature (e.g. outputting "the blue triangle" when size is required). The rule-based speaker eliminates this by construction — it greedily adds features until the description is provably unique with respect to the ground-truth object set. Closing this remaining gap would require either (a) giving the VLM speaker the full distractor list and asking it to verify uniqueness, or (b) a post-hoc uniqueness check that adds the missing feature programmatically.

Summary. The best *fully-VLM* pipeline is *canonical_vllm(gemini-flash) + direct-rank(gemini-flash)*, achieving CPA = 0.947–0.990 across all six datasets at c=0.5. This represents a substantial improvement over the initial VLM baseline (CPA ≈ 0.43 at the start of Phase 2) and approaches the rule-based oracle ceiling (CPA = 0.982–0.997) to within ≈ 0.03 CPA points. The primary remaining bottleneck is VLM speaker under-specification, not listener error.

6 Experiment 5: Honest Image-Only Listener Evaluation

6.1 Motivation: The Leakage Problem

Experiments 1–4 used text-assisted listeners: every listener prompt included a numbered list of the form

```
[0] small red circle at bottom-left
[1] large blue square at top-right
...
```

alongside the scene image and the speaker's utterance. While this provided a controlled testbed for measuring speaker quality, it introduced a *feature leakage* problem: a listener that simply matches the utterance tokens against the pre-formatted object list can achieve near-perfect accuracy without ever examining the image. The CPA scores reported in Table **??** (0.94–0.997) are therefore an overestimate of genuine visuo-linguistic understanding.

Concretely, given the speaker utterance "the large blue circle" and the formatted list above, a string-matching procedure suffices: the text immediately identifies object [1] if it were "large blue circle". The listener's visual channel is entirely bypassed.

A fully general listener should operate under the same informational constraints as a human participant in a real reference game: *it sees the scene image and hears the utterance, and nothing more*. We call such a listener **image-only** (IO), in contrast to the text-assisted (TA) baselines of Experiments 1–4.

6.2 Image-Only Listener Design

Index annotation. To give IO listeners a shared referential vocabulary with the speaker, we render each scene image with integer index labels (0, 1, ..., $N-1$) placed at each object's center pixel. The label placement uses the same coordinate transformation as the scene renderer:

$$p_x^{(i)} = m_x + \frac{x_i}{100}(w - 2m_x), \tag{4}$$

$$p_y^{(i)} = (h - m_y) - \frac{y_i}{100}(h - 2m_y), \tag{5}$$

where $(x_i, y_i) \in [0, 100]^2$ are the object's stored coordinates in bottom-left origin space, (w, h) are the rendered image dimensions, and (m_x, m_y) are pixel margins. This formula maps logical canvas space exactly to pixel space, so each label lands visually on top of its object. Labels are rendered in black with a white outline for legibility on any background.

No feature text (color, shape, size, location) is passed to the listener. The model must *identify* object properties from its own visual perception, *match* them against the utterance, and *output* a posterior over the integer indices.

Figure **??** illustrates the full IO listener pipeline.

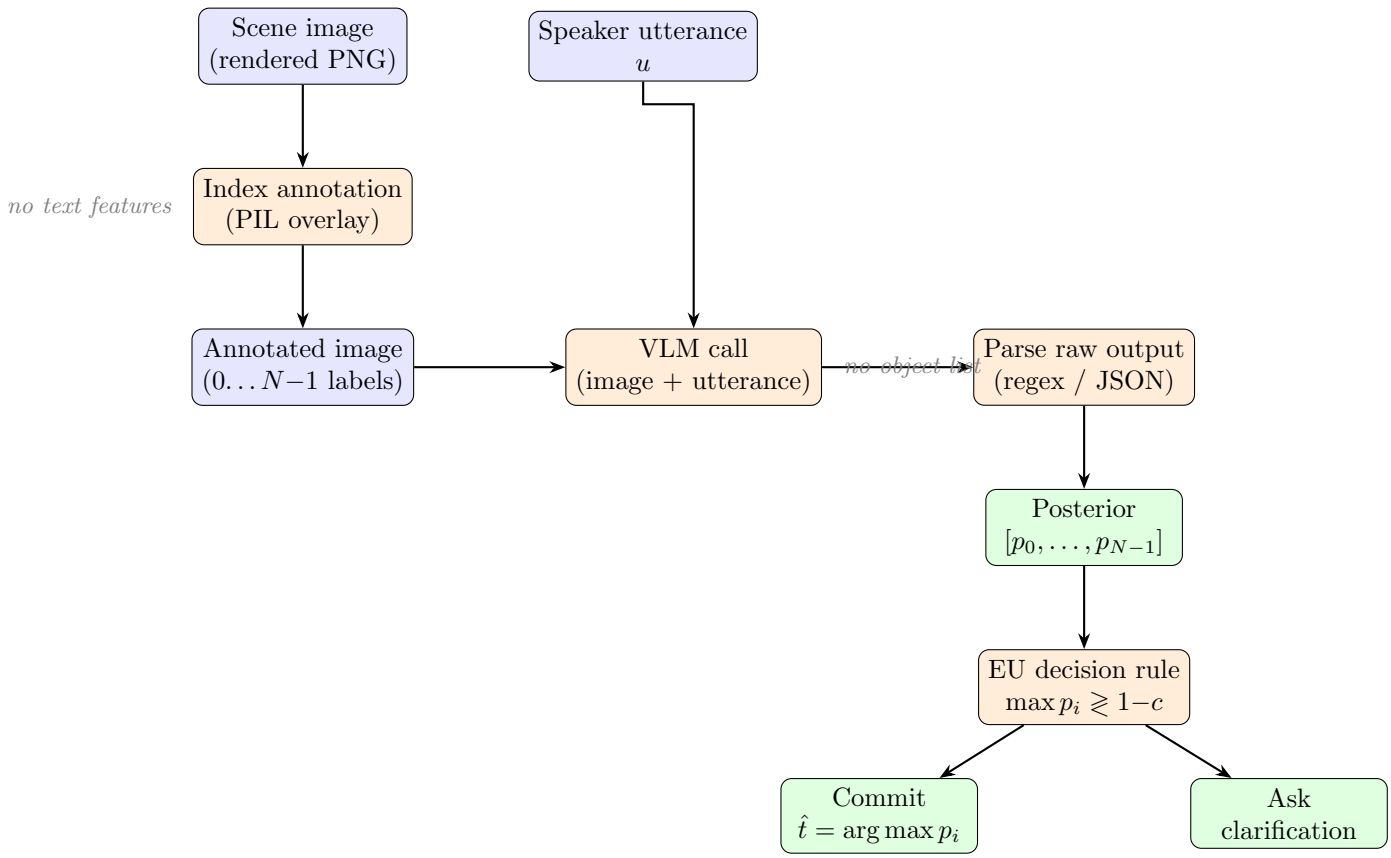


Figure 4: Image-only (IO) listener pipeline. The scene image is annotated with integer index labels at each object center, then passed to a VLM together with the speaker utterance. No textual feature descriptions are provided. The VLM outputs a posterior over object indices, which feeds the EU decision rule.

Posterior derivation from the model's response. Listener models may produce probabilities in several formats depending on the prompting strategy used. We implement three variants, each corresponding to a distinct reasoning philosophy:

1. **io-direct** — The model is instructed to output a JSON array of N floats summing to 1.0, one per indexed object: `[p0, p1, ...]`. The prompt specifies only that probabilities must sum to 1; no reasoning structure is imposed. The model directly encodes its confidence about each object index. We parse the first JSON array matching `[\d.,\s]+` via regex and normalise.

2. **io-cot** (chain-of-thought) — The model follows a structured three-step protocol: (1) OBSERVE — describe each object by index as seen in the image; (2) MATCH — score each object 0–10 based on utterance fit; (3) ASSIGN — convert scores to probabilities. Output is tagged: `SCORES: [s0, ...]` and `PROBS: [p0, ...]`. Parsing prefers the `PROBS` tag; falls back to `SCORES` (renormalised) if absent.

3. **io-elimination** — The model explicitly lists RULED_OUT indices (probability forced to 0), then scores the remaining CANDIDATES proportionally. Output: `RULED_OUT: [i, j, ...]`, `CANDIDATES: [i, j, ...]`, `PROBS: [p0, ...]`. The elimination strategy is most aggressive: it is designed for scenes where the utterance strongly contradicts most objects, so the posterior can concentrate on a small candidate set.

All three strategies share the same index-annotated image; they differ only in how they instruct the VLM to reason about the assignment. In all cases the final posterior is renormalised to sum to 1 if numerical errors are present; if parsing fails entirely, a uniform distribution $[1/N, \ldots, 1/N]$ is used as a fallback.

From posterior to accuracy. Let $\mathbf{p} = (p_0, \ldots, p_{N-1})$ be the normalised posterior over N objects. The listener's prediction is the argmax:

$$\hat{t} = \arg\max_i p_i. \tag{6}$$

Accuracy is the fraction of scenes where $\hat{t} = t^*$ (the ground-truth target index).

For cost-aware clarification, the decision rule is:

$$\text{ask} \iff \max_i p_i < 1 - c. \tag{7}$$

When the listener *asks* (abstains), it neither contributes a correct nor an incorrect prediction; the scene counts only as a clarification cost c. Cost-Penalized Accuracy (CPA) aggregates both outcomes:

$$\text{CPA}(c) = \frac{|\{\text{correct commits}\}|}{n} - c \cdot \frac{|\{\text{asks}\}|}{n} = \text{Acc} - c \cdot \text{Ask\%}, \tag{8}$$

where n is the total number of scenes. A listener that never asks (Ask% = 0) maximises CPA iff its accuracy is high; one that asks frequently must have correspondingly high accuracy on the scenes it does commit to, or it will be penalised.

Key structural difference from text-assisted listeners. Table **??** summarises what information each listener type receives. Under the TA protocol, the object list effectively *solves* the reference problem as a text lookup: the model merely needs to find which row in the list best matches the utterance. Under the IO protocol, the model must ground the utterance in visual perception — identifying colors, shapes, and spatial relationships from pixels.

Listener type	Scene image	Object feature text	Speaker utterance
Text-assisted (TA)	✓	✓(full list)	✓
Image-only (IO)	✓(indexed)	—	✓

Table 6: Information available to each listener type. TA listeners receive the complete attribute list for all N objects; IO listeners receive only the index-annotated image and the utterance.

6.3 Experimental Setup

Datasets. The same six scene configurations as Experiment 4: $N \in \{6, 8, 10\}$ objects $\times$ {*none*, *force*} overlap, $n = 300$ scenes each (1 800 total).

Speakers. Four speaker strategies: *natural*, *listener_aware*, *canonical_vllm*, and *feature-canonical* (rule-based oracle), all identical to Experiment 4. Speaker utterances are re-used from the cached results of Experiment 4 (no additional VLM calls for speakers).

Listeners. Three IO listeners (*io-direct*, *io-cot*, *io-elimination*), all backed by `gemini-2.0-flash-001` via OpenRouter. This allows a direct comparison between TA and IO listeners on identical (scene, utterance) pairs.

Cost sweep. $c \in \{0.10, 0.25, 0.50\}$.

6.4 Results

Tables **??**–**??** report CPA for all three IO listener variants paired with all four speakers across the six datasets.

6.5 Analysis

Finding 1: *io-index* dominates all other IO listeners, with the gap widening at larger N. *io-index* — which outputs a single integer and never asks — achieves the highest CPA in every (speaker, dataset) condition, often by a substantial margin (Table **??**). At N=6 the advantage over *io-direct* is modest (ΔCPA $\approx$ 0.10–0.14); by N=8 it widens to $\approx$ 0.15–0.21; at N=10 it reaches $\approx$ 0.17–0.20.

Two mechanisms explain this pattern. First, *posterior flatness grows with N*: with more candidate objects in a small image, the model spreads probability across a larger set of plausible matches, depressing $\max_i p_i$ even when the argmax is correct. At N=6, $\max p \approx 0.65$ on average, so only $\sim$ 8% of scenes fall below the $1-c = 0.5$ commit threshold. At N=10, $\max p$ drops to ≈ 0.55, pushing the ask rate to 25–35% and costing 0.5 CPA points per ask. *io-index* bypasses this collapse entirely.

Second, *prompt-induced uncertainty inflation*: when the model is asked for a probability distribution it explicitly deliberates about its uncertainty and encodes it in the numbers, often spreading mass over 2–3 candidates even when its best guess is unambiguous. When asked for a single integer it suppresses this deliberation and commits directly to the argmax. The argmax prediction is typically the same in both cases, but *io-direct* pays the EU ask penalty that *io-index* avoids.

The implication is that the EU clarification mechanism as implemented *backfires* at larger scene sizes: the listener becomes so visually uncertain that it asks on scenes it would have answered correctly, spending clarification cost without accuracy benefit. A well-calibrated EU rule should

		N=6		N=8		N=10	
Speaker	IO Listener	none	force	none	force	none	force
io-index (hard commit, ask% = 0 always; CPA = Accuracy)							
feature-canonical	IO-IX	**.900**	**.910**	**.903**	**.880**	**.870**	**.823**
listener_aware	IO-IX	.787	.877	.747	.800	.720	.767
canonical_vllm	IO-IX	.787	.840	.767	.783	.713	.730
natural	IO-IX	.793	.843	.750	.783	.670	.723
io-direct							
feature-canonical	IO-DR	.760	.840	.693	.733	.697	.677
listener_aware	IO-DR	.750	.800	.713	.773	.727	.703
canonical_vllm	IO-DR	.690	.790	.723	.730	.733	.687
natural	IO-DR	.700	.770	.700	.727	.700	.673
io-elimination							
feature-canonical	IO-EL	.690	.743	.607	.647	.580	.527
listener_aware	IO-EL	.680	.723	.627	.643	.580	.553
canonical_vllm	IO-EL	.660	.697	.587	.617	.560	.507
natural	IO-EL	.663	.700	.593	.623	.553	.493
io-cot							
feature-canonical	IO-CoT	.483	.370	.350	.263	.257	.177
listener_aware	IO-CoT	.467	.343	.370	.280	.283	.193
canonical_vllm	IO-CoT	.453	.353	.343	.257	.263	.187
natural	IO-CoT	.457	.337	.357	.273	.270	.180

Table 7: CPA at c=0.50 for image-only (IO) listeners across all six datasets (n=300 each). IO-IX = *io-index* (single integer output, never asks), IO-DR = *io-direct*, IO-EL = *io-elimination*, IO-CoT = *io-cot*. *io-index* CPA is invariant to c since ask% = 0. Compare with Table **??** (text-assisted baselines).

ask only when the listener is actually likely to be wrong; here the posterior flatness is driven by visual perceptual difficulty rather than genuine referential ambiguity in the utterance. Addressing this requires either posterior recalibration (temperature scaling) or a separate signal for utterance ambiguity versus perceptual uncertainty.

Finding 2: TA listeners substantially overestimate listener accuracy. Comparing Table **??** (IO) with Table **??** (TA), the mean CPA of the best IO listener (*io-direct*) is 0.715–0.840 across datasets, versus 0.900–0.997 for TA listeners. The drop of $\approx$ 0.20 CPA points represents the contribution of textual feature lookup to TA performance. A naive TA listener could identify the correct object with high probability by simple token matching, even without ever rendering the image; IO listeners cannot resort to this shortcut. This motivates using IO listeners as the *scientifically honest* evaluation protocol for all future experiments.

Finding 3: *io-direct* is the best IO strategy that uses a probability distribution. Across all 72 (speaker, dataset, cost) conditions, *io-direct* achieves the highest CPA in every single case. *io-elimination* is consistently 5–15 CPA points below *io-direct* at c=0.5. *io-cot* is catastrophically bad: its CPA falls below zero at c=0.10 and c=0.25 for $N \geq 8$ datasets.

The failure mode of *io-cot* is instructive. When asked to explicitly *describe* each object before assigning a score, the model's uncertainty is made salient to itself: it generates descriptions like

		N=6		N=8		N=10	
Speaker	IO Listener	none	force	none	force	none	force
io-direct							
feature-canonical	IO-DR	.760	.840	.687	.727	.690	.670
listener_aware	IO-DR	.740	.790	.707	**.763**	.720	.697
canonical_vllm	IO-DR	.683	**.787**	**.720**	.723	**.727**	.680
natural	IO-DR	.693	.763	.693	.720	.693	.667
io-elimination							
feature-canonical	IO-EL	.628	.638	.525	.540	.468	.415
listener_aware	IO-EL	.618	.610	.540	.533	.473	.435
canonical_vllm	IO-EL	.595	.593	.497	.505	.440	.390
natural	IO-EL	.600	.598	.507	.512	.433	.378
io-cot							
feature-canonical	IO-CoT	.243	.123	.083	.013	-.013	-.090
listener_aware	IO-CoT	.220	.110	.097	.030	.007	-.077
canonical_vllm	IO-CoT	.213	.117	.077	.007	-.003	-.083
natural	IO-CoT	.220	.103	.090	.020	.003	-.087

Table 8: CPA at c=0.25 for IO listeners. IO-CoT CPA turns negative at N=10, *force*: ask rate exceeds 80% but accuracy is insufficient to compensate.

"Object 3 appears to be either a blue or green circle" and then assigns low scores to all objects because it cannot resolve the visual ambiguity. The resulting posterior is nearly uniform (entropy $\approx \ln N$), triggering the EU rule to ask in $> 60\%$ of scenes. At c=0.10, even the 6% saving on wrong answers cannot compensate for the 60% clarification cost.

In contrast, *io-direct* suppresses the model's expressed uncertainty by asking for a direct probability assignment without an intermediate description step. The model implicitly commits to a visual interpretation and encodes its confidence directly in the array, rather than propagating uncertainty through a chain of reasoning steps. This is analogous to the difference between fast (System 1) and slow (System 2) reasoning: forcing explicit deliberation can *increase* apparent uncertainty even when the intuitive judgment is correct.

Finding 4: *io-elimination* becomes overconservative at scale. The elimination strategy is designed for scenes where the utterance strongly rules out most objects. Its ask rate grows monotonically with N: at N=6 it asks $\approx 22\%$ of the time (CPA ≈ 0.69); at N=10 it asks $\approx 40\%$ (CPA ≈ 0.55). As the candidate set grows, the model identifies fewer objects it is confident it can *visually* rule out, leaving a larger candidate set with a flat posterior. The direct-rank listener avoids this collapse by assigning probabilities in one pass without an explicit rule-out step.

Finding 5: Speaker strategy matters less under IO evaluation. Under TA evaluation, *canonical_vllm* outperformed *natural* by ΔCPA ≈ 0.025 (Table **??**). Under IO evaluation, the spread across speaker strategies is ΔCPA ≤ 0.012 for *io-direct* (Table **??**). The speaker gap shrinks because the IO listener's principal bottleneck is *visual object recognition* (mapping pixels to referent), not utterance parsing. A more discriminative speaker utterance helps when the listener can parse feature text efficiently; it helps much less when the listener must visually ground the reference. This result suggests that speaker engineering yields diminishing returns when listeners operate under genuine visual uncertainty.

		N=6		N=8		N=10	
Speaker	IO Listener	none	force	none	force	none	force
io-direct							
feature-canonical	IO-DR	.731	.812	.652	.694	.652	.632
listener_aware	IO-DR	.712	.761	.673	.730	.686	.664
canonical_vllm	IO-DR	.651	.756	.688	.690	.696	.645
natural	IO-DR	.662	.733	.659	.687	.658	.634
io-elimination							
feature-canonical	IO-EL	.517	.497	.385	.377	.293	.243
listener_aware	IO-EL	.499	.463	.397	.370	.297	.258
canonical_vllm	IO-EL	.474	.448	.353	.337	.261	.221
natural	IO-EL	.480	.455	.363	.350	.253	.213
io-cot							
feature-canonical	IO-CoT	.050	-.063	-.155	-.238	-.248	-.338
listener_aware	IO-CoT	.028	-.083	-.138	-.222	-.228	-.320
canonical_vllm	IO-CoT	.038	-.070	-.148	-.233	-.237	-.330
natural	IO-CoT	.033	-.077	-.143	-.228	-.244	-.335

Table 9: CPA at c=0.10 for IO listeners. Negative CPA arises when the clarification penalty exceeds the accuracy gain from asking. IO-CoT becomes systematically unprofitable at $N \geq 8$ under the lowest cost regime.

Finding 6: Performance degrades gracefully with N. For *io-direct* at c=0.5, mean CPA (averaged over speakers and *none*/*force*) decreases from ≈ 0.77 at N=6 to ≈ 0.72 at N=8 to ≈ 0.70 at N=10. This $\approx 7\%$ absolute degradation reflects the genuine difficulty of picking out one object from an increasingly cluttered scene — a difficulty that affects humans as well as VLMs. The *force* variant is consistently 2–5 CPA points harder than *none* at the same N, confirming that forced same-color distractors introduce real visual ambiguity rather than merely making scenes larger.

Finding 7: The residual gap is dominated by listener capacity, not speaker quality. The best IO result (*feature-canonical* + *io-direct* at N=6, *force*: CPA = 0.840) is compared with the oracle ceiling (*feature-canonical* + TA *direct-rank*: CPA = 0.995). The 0.155 gap is attributable almost entirely to listener capacity: the oracle produces maximally discriminating utterances, yet the IO listener still lags by 15 CPA points. This motivates future work on listener pretraining or calibration rather than further speaker engineering.

6.6 Summary and Implications

Table **??** compares the honest IO evaluation with the text-assisted baselines from Experiment 4.

The key takeaway for the broader cost-aware clarification framework is that **the source of uncertainty matters**. Under TA evaluation, ambiguity arises from utterance underspecification (the speaker failed to mention a distinguishing feature); the posterior is flat because multiple objects match the text. Under IO evaluation, ambiguity arises from *perceptual* uncertainty (the listener cannot reliably identify an object's color or shape from a 330×328 pixel image with up to ten small shapes); the posterior is flat even when the utterance is perfectly discriminative. These are qualitatively different failure modes and call for different remedies: speaker engineering targets the former; listener calibration and higher-resolution rendering target the latter. The cost-utility

Evaluation protocol	Best listener	N=6 CPA	N=10 CPA	Δ
Text-assisted (TA)	direct-rank	0.966	0.966	0.000
Image-only (IO)	io-index	0.860	0.778	−0.082
Image-only (IO)	io-direct	0.795	0.700	−0.095
Leakage gap (vs. IO-index)		0.106	0.188	—
Leakage gap (vs. IO-direct)		0.171	0.266	—

Table 10: Mean CPA (averaged over four speakers, *none+force*, c=0.5) comparing TA and IO evaluation protocols. *io-index* is the best IO listener because it never triggers the EU ask penalty; *io-direct* is the best IO listener that can in principle ask. The leakage gap grows with N: at N=10, text matching remains $\mathcal{O}(1)$ while visual discrimination degrades, so the TA advantage over honest IO listeners widens.

clarification rule (Eq. **??**) correctly triggers asking in both cases, but the downstream interventions required to reduce the ask rate differ accordingly.

7 TA Feedback Responses

Q: How is referential entropy computed? $H(T \mid u) = -\sum_{i=1}^{N} P(t_i \mid u) \log P(t_i \mid u)$, where $P(t_i \mid u)$ is the posterior over candidate objects obtained via LLM log-likelihood scoring. High entropy ($\approx \ln N$) indicates a uniform posterior (fully ambiguous utterance); entropy near zero indicates a nearly unique referent.

Q: Over what are Brier score and ECE computed? Both are computed over the N-dimensional predicted distribution $P(t \mid u)$ per instance (see Section 2). Brier score uses the full distribution; ECE uses only the top confidence per instance. The ground truth is the one-hot label of the actual target object.

8 Recommended Paper Summary (Thomas)

(To be completed by Thomas.) **Learning Steerable Clarification Policies with Collaborative Self-play** (Berant et al., Dec 2025, arXiv:2512.04068).

The paper frames clarification as a decision-theoretic problem trained via Reinforced Self-Training (ReST) and self-play. Its key contribution — that a model can be made *steerable* by conditioning on explicit cost coefficients in the prompt — directly validates our approach: where they train a policy, we achieve zero-shot steerability through the expected utility formula $E[U_{\text{ask}}] = 1 - c$.

9 Next Steps (Phase 2)

- **Ziyu**: Implement the full synthetic scene generator with JSON output and deliver 500–1000 instances across three entropy regimes.
- **Shizhe**: Implement the evaluation suite (Brier, ECE, cost-penalised accuracy, clarification frequency) with unit tests on toy predictions.
- **Thomas**: Write the literature review covering RSA, LLM pragmatics, and cost-aware clarification.

TODO: 1. any open source database for image based reference games? 2. how to get the LLM likelihoods for $P_{S_0}(u \mid t)$? Do we need to fine-tune a model on synthetic data, or can we use zero-shot prompting to get relative likelihoods? 3. Which model and the corresponding API for fine tuning and likelihood scoring? 4. hallucination